

Part A: Fundamentals – Q&A

Q1. What is Data Analysis?

Data Analysis is the process of inspecting, cleaning, and transforming raw data to discover useful patterns, trends, and insights that support decision-making.

Q2. How do you plan a Data Science Project? (Steps)

1. Problem Definition – understand the business goal.
2. Data Collection – gather data from relevant sources.
3. Data Cleaning – handle missing values, duplicates, errors.
4. Exploratory Data Analysis (EDA) – visualize and understand patterns.
5. Feature Engineering – create/select relevant variables.
6. Model Building – apply ML algorithms.
7. Evaluation – test model performance with metrics.
8. Deployment & Monitoring – put model into use and track it.

Q3. Frame a Machine Learning problem statement for predicting churn.

Problem: Given a customer's demographic and purchase behavior data (tenure, contract type, charges, services used), predict whether the customer will churn (Yes/No).

Type: Supervised, Binary Classification.

Input (X): customer attributes (age, income, tenure, contract, charges, etc.)

Output (y): Churn – Yes / No.

Goal: Identify at-risk customers early so the business can act to retain them.

Q4a. What are Tensors?

A tensor is a generalized container for numbers that can have any number of dimensions. A scalar is a 0-D tensor, a vector is 1-D, a matrix is 2-D, and higher-dimensional data (e.g., a batch of color images) form 3-D, 4-D tensors and beyond. In ML, tensors are how data, weights, and gradients are stored and processed.

Q4b. Explain tensors with simple NumPy examples.

```
import numpy as np

scalar = np.array(5) # 0-D tensor
vector = np.array([1, 2, 3]) # 1-D tensor
matrix = np.array([[1,2],[3,4]]) # 2-D tensor
tensor3d = np.random.rand(2,3,3) # 3-D tensor

print(scalar.ndim) # 0
print(vector.ndim) # 1
print(matrix.shape) # (2, 2)
print(tensor3d.shape)# (2, 3, 3)
```

Here, **ndim** tells the number of dimensions and **shape** tells the size along each dimension — e.g. a batch of 2 RGB images of size 3x3 would be a 4-D tensor of shape (2, 3, 3, 3).